Zone Structure of Sudoku: Cross-Box Invariants, the 135 Rule, and Structure-Preserving Puzzle Generation

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March 2026

Abstract

We introduce a structural decomposition of completed Sudoku grids based on *zones*: sets of nine cells occupying the same relative position within each 3×3 box. This viewpoint yields a simple conserved-sum law, which we call the *135 Rule*: the zone sums along any template row or column equal exactly 135. We also define the *zone-pair overlap vector*, a 36-dimensional descriptor recording the number of shared distinct digits for each pair of zones, and prove that this vector is invariant under global digit permutation.

We then report computational evidence that, for several fixed clue masks, the zone-pair overlap vector acts as a strong discriminator between unique and multi-solution instances. In experiments on three seed masks (17, 22, and 30 clues), all 300 digit-permuted unique instances shared the same overlap vector, while none of 1,500 sampled solutions drawn from multi-solution instances matched that vector. Building on these observations, we describe *LarsForge*, a structure-preserving puzzle generation pipeline based on uniqueness-preserving symmetries and digit permutations, and a zone-guided grid construction procedure that we found to be effective in practice.

Our main contribution is not a proof of polynomial-time uniqueness verification, but a new cross-box coordinate system for Sudoku together with initial theoretical and experimental evidence that it captures structural information not explicit in the classical row-column-box formulation.

1 Introduction

Sudoku is a finite-domain constraint satisfaction problem on a 9×9 grid partitioned into nine 3×3 boxes. A completed grid is valid if each row, column, and box contains the digits 1–9 exactly once. The generalized problem is NP-complete [1], and the minimum number of clues for a uniquely solvable standard Sudoku puzzle is 17 [2].

Most published analyses of Sudoku emphasize the row-column-box decomposition. That viewpoint is natural and fundamental, but it is not the only way to group the 81 cells. In this paper we study a complementary decomposition, which we call the *zone structure*. A zone consists of all cells occupying the same relative position inside their respective 3×3 boxes. This creates nine cross-box cell classes, each of size nine, and induces a secondary 3×3 "template grid" of zone statistics.

This perspective leads to two main observations. First, the zone sums satisfy a simple conservation law: every template row and template column sums to 135. Second, certain zone-derived descriptors are unchanged by global digit relabeling. These descriptors can therefore be used to compare puzzles that share a clue mask but differ by digit permutation.

The present paper has two goals. The first is formal: to define zones precisely, prove the 135 Rule, and record several immediate consequences. The second is computational: to test whether zone-derived descriptors are useful in practice for distinguishing unique from multi-solution instances under fixed masks, and to describe how these descriptors can be incorporated into a structure-preserving generation pipeline.

Our contributions are as follows.

(1) We formally define Sudoku zones and prove the 135 Rule (Section 2).

(2) We define the zone-pair overlap vector and prove its invariance under global digit permutation (Section 3).

(3) We report experiments on three clue masks in which the overlap vector perfectly separated our sampled unique and multi-solution instances (Section 4).

(4) We describe LarsForge, a generation pipeline built from uniqueness-preserving transformations of verified seed puzzles (Section 5).

(5) We describe a zone-guided grid construction strategy and summarize its observed runtime in our implementation (Section 6).

All code, puzzle collections, and related tooling are available at `https://github.com/oppressionslayer/wsrf-sudoku-solved-series`, with installable packages including `larsdoku` and `larsdoku-zs`.

2 Zone Structure and the 135 Rule

Definition 1 (Zone)**.** *A* zone *is the set of nine cells occupying the same relative position within each of the nine* 3×3 *boxes. There are nine zones, corresponding to the nine positions within a box:*

$$\mathrm{TL},\ \mathrm{TC},\ \mathrm{TR},\ \mathrm{ML},\ \mathrm{MC},\ \mathrm{MR},\ \mathrm{BL},\ \mathrm{BC},\ \mathrm{BR}.$$

Each cell belongs to exactly one zone, and the nine zones partition the 81 cells into nine disjoint sets of size nine.

Definition 2 (Zone sum)**.** *For a completed Sudoku grid, the* zone sum Z_i *is the sum of the nine digits in zone* i*.*

Definition 3 (Template grid, template rows, and template columns)**.** *The nine zone sums can be arranged in a* 3×3 *template grid:*

$$\begin{pmatrix} Z_{\mathrm{TL}} & Z_{\mathrm{TC}} & Z_{\mathrm{TR}} \\ Z_{\mathrm{ML}} & Z_{\mathrm{MC}} & Z_{\mathrm{MR}} \\ Z_{\mathrm{BL}} & Z_{\mathrm{BC}} & Z_{\mathrm{BR}} \end{pmatrix}.$$

A template row *is a row of this grid, and a* template column *is a column of this grid.*

Theorem 4 (135 Rule)**.** *In any valid completed Sudoku grid, the zone sums along any template row or any template column add to 135.*

Proof. Consider the template row $(\mathrm{TL}, \mathrm{TC}, \mathrm{TR})$. These three zones collect the top-left, top-center, and top-right cells from each of the nine boxes. Across the full grid, those 27 cells are exactly the cells in the first three rows of the Sudoku.

Each row of a valid completed Sudoku contains the digits 1–9 and therefore has sum 45. Hence the first three rows sum to $3 \cdot 45 = 135$, which proves

$$Z_{\mathrm{TL}} + Z_{\mathrm{TC}} + Z_{\mathrm{TR}} = 135.$$

The same argument applies to the other two template rows. By the analogous box layout argument in the orthogonal direction, the same conclusion holds for each template column. □

Corollary 5. *The nine zone sums satisfy five independent linear constraints given by the three template-row equations and three template-column equations, minus one redundancy coming from equality of the total row sum and total column sum. Equivalently, the 9-dimensional zone-sum vector has 4 degrees of freedom.*

3 Zone-Pair Overlaps as Permutation Invariants

Definition 6 (Zone-pair overlap)**.** *Let A and B be two zones in a completed grid. Define*

$$\omega(A, B) = |D(A) \cap D(B)|,$$

where $D(Z)$ is the set of distinct digits that appear in zone Z.

Since there are $\binom{9}{2} = 36$ unordered pairs of zones, each completed grid induces a 36-dimensional *zone-pair overlap vector* $\boldsymbol{\omega}$.

Proposition 7 (Invariance under digit permutation)**.** *Let $\pi : \{1, \ldots, 9\} \to \{1, \ldots, 9\}$ be a bijection, and let G' be the grid obtained from a completed grid G by replacing each digit d by $\pi(d)$. Then for every pair of zones (A, B),*

$$\omega_{G'}(A, B) = \omega_G(A, B).$$

Proof. Because π is bijective, the set of digits appearing in zone A is mapped to $\pi(D(A))$, and similarly for B. Therefore

$$|D'(A) \cap D'(B)| = |\pi(D(A)) \cap \pi(D(B))| = |D(A) \cap D(B)|.$$

Thus the overlap cardinality is preserved. □

Definition 8 (Lock pattern)**.** *For a fixed clue mask M, define the* lock pattern *of a uniquely solved instance with mask M to be its zone-pair overlap vector. By Proposition 7, any global digit permutation of that puzzle has the same lock pattern.*

4 Empirical Uniqueness Discrimination via Zone Locks

We tested whether the overlap vector is useful for distinguishing unique from multi-solution instances that share the same clue mask.

4.1 Experimental setup

For each of three seed masks (17, 22, and 30 clues), we performed the following experiment:

(1) Starting from a verified unique seed puzzle, generate 100 unique instances by global digit permutation.

(2) Generate 100 multi-solution instances with the same clue mask by assigning random valid digits to the clue positions.

(3) For each multi-solution instance, obtain 5 solutions using a randomized backtracking solver.

(4) Compute the zone-pair overlap vector for every sampled solution.

(5) Check whether any sampled multi-solution vector matches the lock pattern of the corresponding unique seed family.

4.2 Results

Seed mask	Unique tested	Multi solutions tested	Multi matching lock
17 clues	100/100	500	0 (0.0%)
22 clues	100/100	500	0 (0.0%)
30 clues	100/100	500	0 (0.0%)
Total	**300/300**	**1,500**	**0**

Table 1: Zone-lock experiment. In these experiments, all 36 overlap entries were constant across digit-permuted unique instances, and none of the sampled multi-solution instances matched the corresponding lock pattern.

Across these experiments, the overlap vector perfectly separated the sampled unique and multi-solution families. We therefore view the lock pattern as a promising *empirical necessary condition* for uniqueness under a fixed mask. We do *not* claim sufficiency here.

4.3 Higher-order zone structure

Beyond the 36 overlap coordinates, we observed additional regularities in joint conditions on zone sums.

Zone-sum correlations (seesaw effect). The 135 Rule imposes a conservation law that creates strong negative correlations between zone sums along shared template rows and columns. In a sample of 500 puzzles from the forum hardest SE 11+ collection, we measured the following pairwise Pearson correlations among zone sums:

Every one of the ten strongest correlations is negative and connects zones that share a template row or column. No positive correlation exceeds $r = 0.33$. This seesaw structure means that the zone sums are not independent descriptors but form a tightly coupled constraint network.

Zone pair	Correlation r	Relationship
TL – BL	-0.574	same template column
TL – TR	-0.574	same template row
MC – BC	-0.559	same template column
TR – BR	-0.539	same template column
BL – BR	-0.537	same template row
ML – MR	-0.520	same template row

Table 2: Strongest zone-sum correlations. All are negative and occur between zones sharing a template row or column, consistent with the seesaw effect of the 135 Rule: when one zone sum increases, its template-row or template-column partner must decrease.

Triple conditioning and digit crystallization. Conditioning on two or three zone sums simultaneously produces much stronger constraints. For example, conditioning on $Z_{\mathrm{MC}} = 51$, $Z_{\mathrm{TL}} = 48$, and $Z_{\mathrm{ML}} = 47$ in a sample of 2,000 23-clue puzzles produced a family in which 43 of 81 digit-cell assignments were fixed at 100% across all sampled instances. With only a single zone sum conditioned, no digit-cell assignment reached 100% presence. This crystallization effect suggests that the zone-sum vector encodes substantial structural information that becomes visible only through joint conditioning.

These observations are preliminary, but they suggest that the zone-sum vector and overlap vector may jointly encode stronger structural information than either statistic alone.

5 LarsForge: Structure-Preserving Puzzle Generation

LarsForge is a puzzle-generation pipeline built from transformations that preserve Sudoku validity and, when applied to a verified unique seed, preserve uniqueness as well.

5.1 Digit-permutation forge

Applying any global permutation of the digits 1–9 to a verified unique seed produces another verified unique puzzle with the same clue mask. This yields

$$9! = 362{,}880$$

instances per seed.

5.2 Shuffle forge

Additional structure-preserving transformations include row permutations within bands, column permutations within stacks, band permutations, stack permutations, and transposition. Their combined action yields up to

$$362{,}880 \times 3{,}359{,}232 \approx 1.22 \times 10^{12}$$

instances per seed.

5.3 Constellation families

For certain seed families, especially masks in the 23–24 clue range, we observed a striking property: *every* tested digit permutation of a verified unique seed remained unique. We tested this exhaustively on several masks, verifying all $9! = 362{,}880$ permutations, and found 100% uniqueness preservation. We refer to such families as *constellations*.

For constellation families, the full 9! orbit can be generated with *zero* uniqueness verification overhead beyond the original seed. This is significant because uniqueness checking is typically the computational bottleneck in puzzle generation. The constellation property effectively eliminates this bottleneck for the affected mask families.

We verified the constellation property on masks with 23–24 clues and observed it consistently. Whether this property holds for all well-formed masks with sufficient clue density remains an open question.

5.4 Performance

Using a seed bank of 1,039 puzzles spanning 17–27 clues, LarsForge generates puzzles at 1.5×10^5 puzzles per second in pure Python (CPython 3.13, single core, no C extensions beyond NumPy and Numba for the solver). All performance numbers reported here are reproducible using the open-source `larsdoku` package.

A pre-computed database of 93,438 completed grids, indexed by zone-sum pattern, supports grid retrieval and digit-permutation generation at $6.5\,\mu$s per grid ($\approx$ 153,000 grids per second). Combined with the full Sudoku symmetry group ($\approx 1.22 \times 10^{12}$ transformations per seed), the database yields over 10^{17} distinct valid grids.

Operation	Rate	Method
Grid retrieval + digit permutation	$6.5\,\mu$s/grid	database lookup
Puzzle generation (digit permutation)	150,000/s	seed bank
Puzzle generation (full shuffle)	$\sim$ 100,000/s	shuffle + permute
Grid construction from zone sums	~ 1 ms	zone-incremental DFS

Table 3: Measured generation rates in our Python implementation.

6 Zone-Guided Grid Construction

We also implemented a zone-guided grid construction strategy that attempts to build valid completed grids directly from zone-sum specifications satisfying Theorem 4.

6.1 High-level procedure

Given target zone sums $Z_0, \ldots, Z_8$ satisfying the 135 Rule, the procedure is:

(1) order the zones by heuristic constraint density,

(2) fill each zone using constraint propagation while maintaining row/column/box compatibility and the target zone sum, and

(3) continue until a completed valid grid is obtained.

In our experiments this zone-incremental construction often completed in about 1 ms in Python. Because the implementation is heuristic and engineering-heavy, we present this result here as a computational observation rather than as a new combinatorial theorem.

7 Discussion

The zone structure provides a cross-box coordinate system for Sudoku that is complementary to the standard row-column-box view. Classical constraints act within rows, columns, and boxes; zone statistics aggregate information across all nine boxes at fixed relative positions.

The main mathematical results of this paper are modest but clean: the 135 Rule and permutation invariance of the zone-pair overlap vector. The more ambitious claims are computational and should be interpreted accordingly. In particular, our experiments indicate that zone locks are highly informative for the tested masks, but they do not yet establish a complete characterization of uniqueness.

Two research directions seem especially natural. First, one can ask whether the zone-pair overlap vector, possibly together with the zone-sum vector, yields a sharper necessary condition or even a sufficient condition for uniqueness under restricted families of masks. Second, one can augment algebraic encodings of Sudoku with zone-derived constraints and test whether this reduces null-space dimension in practice.

8 Availability

All software, puzzle collections, and related data used in this project are publicly available:

- **Solver and forge:** `pip install larsdoku`
- **Zone solver tools:** `pip install larsdoku-zs`
- **Source and solved series:** `https://github.com/oppressionslayer/wsrf-sudoku-solved-series`
- **Documentation:** `https://larsdoku-docs.netlify.app/`

References

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